

(a) Conditions + structure

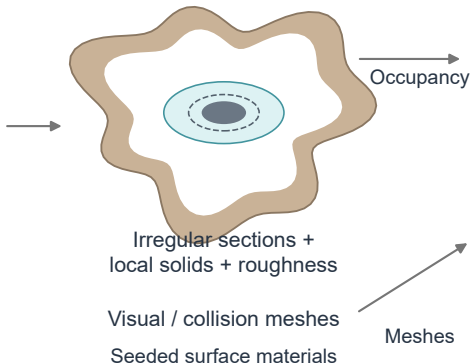
Sample S/G coordinates

Config / seed
Robot radius + margin



Routes + chambers

(b) Protected geometry



(c) Task verification

S/G + radius

Independent 3D A*

Both final meshes
Inside + clearance

Failures retained

(d) Delivery

Accept

Textured assets
+ task packs

Optional portals / export / props

Check changed surfaces; separate records

Protected generation; checked polylines

$$\rho_g = r + m + 2.1\Delta_c$$

$$\hat{d}_{\min} - h/2 - \varepsilon > r + m$$

Solid teal: construction Dashed violet: searched path